

Docking Universal - Preliminary Pocket-Review Report (Not a Protocol)

Report generator version 0.7.0

Target: 2R8N

Interpretive status: candidate-site evidence for user review; no docking site has been selected or approved

1. Candidate pocket review

This preliminary report presents fpocket candidates and any qualifying related-structure ligand evidence. Automatic selection will use the top-ranked fpocket box. This report does not authorize screening.

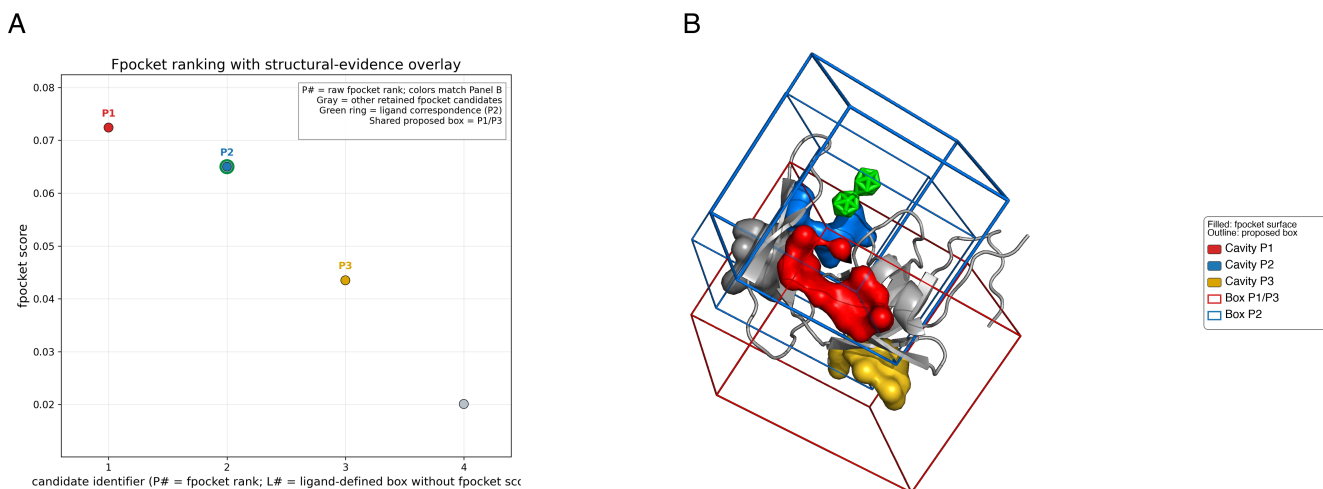
Pocket configuration	Recorded value
Interpretive status	Pocket review only; not a protocol
fpocket eligibility threshold used	0.0 (relaxed sensitivity pass)
Top fpocket score	0.0724
Top-pocket druggability descriptor	0.0
Selected docking region	Not yet selected; choose one or more boxes after reviewing this report

Related-structure ligand evidence

Deposited ligands from the same protein, exact-sequence structures, and explicitly approved close structural homologs are treated as supporting ligand-location evidence. In the source lists, exact match means the same protein identifier or 100% sequence identity; partial match means an approved close structural homolog meeting the recorded thresholds. These sources do not independently validate a binding site or protocol, and orthology is not inferred. Ligands were aligned by their contacting protein chain and grouped by spatial overlap. Direct fpocket correspondence requires unique cavity contact and full ligand containment. Boxes start at 26 Å and expand within the recorded 36 Å dimension and 64,000 Å³ volume limits. P# denotes an fpocket-derived box; L# denotes ligand evidence without a matching cavity. Accepted alignments require identity 0.9, coverage 0.7, and C-alpha RMSD at most 2.0 Å. Ambiguous chain assignments were excluded. Excluded ambiguous chain assignments: 1ZTZ/CB5/P/1001.

Region	Evidence-based box recommendation
P2	Box: 26.0 × 26.0 × 26.0 Å; Direct support (fully contained by P2): Partial matches (close structural homolog): 1ZTZ/CB5, 3I8W/CB5. Additional ligand evidence at the same aligned site (supports the site location; not used to set Box P2 dimensions): Exact matches (100% sequence identity): 2R5P/MK1, 2R5Q/1UN; Partial matches (close structural homolog): 1D4Y/TPV, 1HPO/UNI, 1MRW/K57, 1MSM/JE2, 2PK5/O75, 2PK6/O33, 3KDB/O06, 3KDC/JZP, 3KDD/JZQ, 3NLS/O16, 3QBF/JHG.

Evidence review. Related-structure ligand locations were compared with fpocket candidates. The standard fpocket threshold did not provide sufficient correspondence, so candidates from a relaxed-sensitivity pass were retained. No docking box was selected in this preliminary report. No close structural-homolog evidence was used.



Selectable docking boxes

Automatic selection uses the top-ranked fpocket box. These alternatives are available by choosing reviewed selection.

Selectable box	Evidence basis
P1	Individual fpocket cavity box. No direct correspondence in the retained deposited-ligand evidence.
P2	Individual fpocket cavity box; direct deposited-ligand correspondence recorded. Direct deposited-ligand correspondence: Partial matches (close structural homolog): 1ZTZ/CB5, 3I8W/CB5. Additional ligand evidence at the same aligned site (supports the site location; not used to set this box's dimensions): Exact matches (100% sequence identity): 2R5P/MK1, 2R5Q/1UN; Partial matches (close structural homolog): 1D4Y/TPV, 1HPO/UNI, 1MRW/K57, 1MSM/JE2, 2PK5/075, 2PK6/O33, 3KDB/006, 3KDC/JZP, 3KDD/JZQ, 3NLS/O16, 3QBF/JHG.
P3	Individual fpocket cavity box. No direct correspondence in the retained deposited-ligand evidence.
P4	Individual fpocket cavity box. No direct correspondence in the retained deposited-ligand evidence.
P1/P3	Consolidated box spanning the named overlapping fpocket cavities. No direct correspondence in the retained deposited-ligand evidence.

2. Reproducibility, software, and references

fpocket generated the cavity candidates and descriptors. Receptor preparation: strict Meeko succeeded; PDBFixer was not needed. No docking or interaction analysis was performed. PyMOL, Matplotlib, and ReportLab generated this report.

Software versions used for this report

Result element	Software	Version used
Scientific workflow	Docking Universal	0.7.0
Ligand-free cavity detection	fpocket	4.2.2
Receptor preparation	Meeko	0.7.1
3D rendering	PyMOL	3.0.0
Plots	Matplotlib	3.9.4
PDF generation	ReportLab	5.0.0
Scientific workflow runtime	Python	3.9.23

Scientific and software references

1. Le Guilloux V, Schmidtke P, Tuffery P. Fpocket: an open source platform for ligand pocket detection. BMC Bioinformatics. 2009;10:168. <https://doi.org/10.1186/1471-2105-10-168>
2. Santos-Martins D, He Y, Eberhardt J, et al. Meeko: molecule parameterization and software interoperability for docking and beyond. J Chem Inf Model. 2025;65:13045-13050. <https://doi.org/10.1021/acs.jcim.5c02271>
3. The PyMOL Molecular Graphics System, Version 3.0.0, Schrodinger, LLC. <https://www.pymol.org/support.html>
4. Hunter JD. Matplotlib: A 2D Graphics Environment. Comput Sci Eng. 2007;9:90-95. <https://doi.org/10.1109/MCSE.2007.55>
5. ReportLab: PDF generation toolkit and user guide. <https://docs.reportlab.com/>
6. Python Software Foundation. Python Language Reference. <https://www.python.org/>